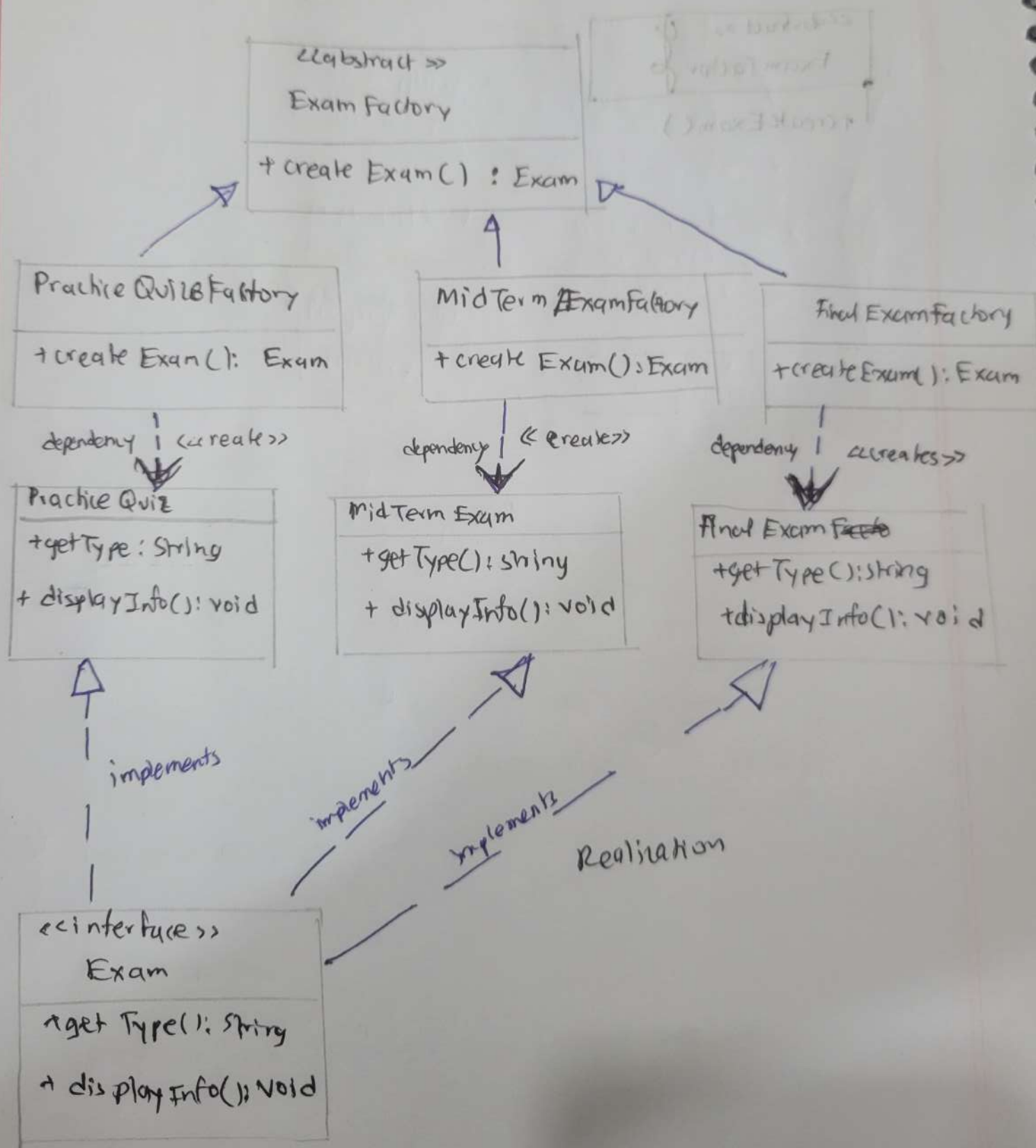


PART A A T 9/19



PART B

<<interface>>

QuestionFactory

+ createQuestion(): Question
+ createRenderer(): QuestionRenderer
+ createEvaluator(): QuestionEvaluator

Essay Factory

- source: QuestionSource
+ createQuestion(): Question
+ createRenderer(): QuestionRenderer
+ createEvaluator(): QuestionEvaluator

<<creates>>

<<creates>>

creates

Essay Question

- text: String
- points: int
- difficulty: String
+ getType(): String
+ getPoints(): int
+ getDifficulty(): String

EssayRenderer

render
(question: Question)
: void

Essay Evaluator

+ evaluate (QuestionAnswer):
int

<<interface>>

Question

+ getType(): String
+ getPoints(): int
+ getDifficulty(): String

<<interface>>

QuestionRenderer

render (Question:
void)

Question Evaluator

+ evaluate (Question,
Answer): int

PART 8

Similarly

creates

creates

Question
Factory

~~MCQ~~ MCQ
factory

T/F Factory

Programming
Factory

MCQ
Evaluator

MCQ
Renderer

MCQ
Question

TF eval

TF Rend

TF Question

Programming Evaluator

Programming Renderer

Programming Question

interface
Question

interface
Question
Renderer

interface
Question
Evaluator

~~Question~~

670v

def __init__(self):
 self.questions = []
 self.renderers = []
 self.evaluators = []
 self.questions.append(
 MCQQuestion(
 "What is the capital of France?",
 ["Paris", "London", "Berlin", "Madrid"],
 "Paris"
))
 self.renderers.append(
 MCQRenderer()
)
 self.evaluators.append(
 MCQEvaluator()
)

PART C

Exam Configuration

- title: String
- duration: int
- passing score: int
- neg mark: boolean
- ques shuffle: boolean
- auto submit: boolean
- calculator allowed: boolean

- + getTitle(): String
- + getDuration(): int
- + getPassingScore(): int
- + hasNegMarking(): boolean
- + hasQuestionShuffle(): boolean
- + hasAutoSubmit(): boolean
- + isCalculatorAllowed(): boolean
- + displayConfig(): void

Exam Builder

- title: String
- duration: int
- passing score: int
- neg marking: boolean
- question shuffle: boolean
- auto submit: boolean
- calculator allowed: boolean

- + getTitle(): ExamBuilder
- + getDuration(): ExamBuilder
- + getPassingScore(): ExamBuilder
- + hasQuestionShuffle(): ExamBuilder
- + hasAutoSubmit(): ExamBuilder
- + isCalculatorAllowed(): ExamBuilder
- + displayConfig(): void
- + build(): ExamConfiguration

ExamConfiguration

Part D

«interface»
Question Source

+ get Question (type: string) : Question

+ get source Name () : String

Question Bank Source

- question Bank : Question Bank

+ get Question (Type: string) : Question

+ get source Name () : String

Manual Question Source

+ get Question (type: string) : Question

+ get source Name () : String

Fake Question Source

- faker : Question Faker

+ get Question (type: string) : Question

+ get source Name () : String

«interface»
Question

+ get Type () : String

+ get Point () : int

+ get difficulty : String

Question Bank

- random : Random

- MCQ : Question[]

- Essay : Question[]

- program : Question[]

- T/FQ : Question[]

+ get Random Question

(type: string) : Question